

K530 — VERIFICATION Thursday Marathon: Lyra F325 SELF-RETRACTED (Subtle Index-Flow Artifact in Super-Jacobi Encoding; Jacobiator Never-Zero Contradicts F(4) Basic-Classical-Lie-Superalgebra Rigidity Theorem; Lyra Did NOT Force Second κ Number — “Would Be Exactly the Wrong Response”) + F319 Status RESTORED-OPEN (Retraction Based on Now-Retracted F325) + Lyra F326 EXPLICIT GENERATION WAVEFUNCTIONS $\psi_k(z) = (z_1 + iz_2)^k \otimes u_0$ DELIVERED Sympy-Verified Harmonic UNBLOCKING Elie Mass Kernel + Elie REDIRECTION: Aux-Vanishing AUTOMATIC for Any Dirac Square (Clifford Algebra Rank-0 + Rank-2 Only) $\rightarrow$ Aux-Vanishing Necessary But WEAK Doesn’t Distinguish F(4) $\rightarrow \{Q,Q\}=D^2$ Cannot Derive #359 Alone $\rightarrow$ Conformal Sector Decisive (Genuine REDIRECTION of #1 Foundational Effort) + Cal #389 THREE-WAY Aux Framing (Vanish $\rightarrow$ F(4) Exactly; Central-Nonzero $\rightarrow$ Central Extension F(4)-with-Charges Tier-Distinct; Non-Central $\rightarrow$ Fails Bar 2) Distinguishes “F(4)” from “F(4)-with-Charges” + Grace INDEPENDENT κ ANGLE Delivered: F(4) Supertrace Consistency via Dynkin Index Balance ($\mathfrak{so}(7)$ -index = $\mathfrak{sl}(2)$ -index = 4 — Killing-Form Constraint κ Must Respect) — Genuine Cal #35 Two-Route Cross-Check with Lyra Conformal $\{Q,S\}/\{S,S\}$ Closure Route + Lyra DISCIPLINE AT PEAK: Caught Own Bug + Refused Solo Retry + Deferred to Paired Computation + Cal #35 STANDING Applied Retroactively to K529 “Two Proofs of One Finding” Framing — F325 Retraction Means Honest Count Goes Back to ONE Pending Route (Elie 4382 Abstract Argument) Whose Status is ITSELF Now in Question per F(4) Rigidity Theorem; κ Source GENUINELY OPEN Needing Careful Paired Computation NOT Solo Redo

Date: 2026-06-25 (Thursday marathon continued, date-verified post-K529) · Auditor: Keeper · Inputs: Lyra F325 bug-catch retraction + F326 wavefunctions delivered + κ honest open; Elie aux-vanishing automatic redirection; Cal #389 three-way framing; Grace independent κ angle via Dynkin index balance

Verdict — Lyra F325 SELF-RETRACTED (bug caught honestly + no forced second number); F319 RESTORED-OPEN; F326 wavefunctions UNBLOCK Elie mass kernel; Elie aux-vanishing AUTOMATIC for Dirac square $\rightarrow$ REDIRECTION to conformal sector; Cal #389 three-way framing distinguishes F(4) from F(4)-with-charges; Grace independent Dynkin index balance angle for κ ; κ source GENUINELY OPEN needing careful Lyra+Grace paired computation

Count holds 4 of 26. Multiple within-session honest retractions + redirections continuing the discipline pattern. Substantive substrate-architectural item: F(4) basic-classical-rigidity theorem says κ MUST be fixed somewhere — abstract Elie 4382 argument that “ $\{Q,Q\}$ Jacobi doesn’t fix κ ” is now in question; needs careful paired redo.

Landing 1 — Lyra F325 SELF-RETRACTION via bug catch

The honest catch

“I computed κ (F325), posted it, then checked it against a theorem and it failed — my Jacobiator gave a never-zero result, which would mean $F(4)$ ’s bracket doesn’t close. It does ($F(4)$ is a basic classical Lie superalgebra). So that computation is bugged and I pulled it.”

Substantive within-session bug catch: - F325 Clifford computation gave $\|J(\kappa)\|^2 = 2268 + 69\kappa^2$ (never zero for any κ) - Never-zero Jacobiator contradicts $F(4)$ bracket closure - $F(4)$ IS a basic classical Lie superalgebra (theorem) — bracket DOES close - Contradiction → computation BUGGED → RETRACTED

Per [[feedback_quaker_method]] + [[feedback_engage_dont_label_compute_beats_calibrate]] + Cal #347 / B5 STANDING: Lyra cross-checked own result against external theorem; found contradiction; retracted. Substantive discipline at peak.

Per Lyra discipline at peak — refused solo retry

“I attempted the fix just now; the bug is a subtle index-flow artifact in the super-Jacobi encoding, not a one-line normalization. I’m deliberately not forcing a second number. The correct κ is exactly the thing the Grace-pairing was set up for — a careful joint rep-theory computation — and that’s where it belongs, not in a rushed solo retry right after a bug.”

Per [[feedback_target_innocence_lens_derived_vs_fit_discipline]] + [[feedback_engage_dont_label_compute_beats_calibrate]] + Cal #347 / B5 STANDING: Lyra refused to force second number after bug catch. Substantively right. Solo retry immediately after caught error has high risk of producing same-class error with confirmation bias.

Substantive substrate-architectural fallout

Per $F(4)$ rigidity theorem: if $F(4)$ is basic classical Lie superalgebra with definite even+odd structure, then κ (relative coefficient between $\mathfrak{so}(7)$ and $\mathfrak{sl}(2)_R$ in $\{Q,Q\}$) IS uniquely determined by Jacobi. The “ $\{Q,Q\}$ Jacobi doesn’t fix κ ” finding from Elie 4382 + Lyra F325 must be WRONG somewhere.

Possibilities: 1. Elie 4382 abstract argument has a flaw (disjoint tensor indices argument missing structural constraint) 2. Both K-audits were computing some different bracket than the $F(4)$ supercharge bracket 3. There’s a normalization/encoding subtlety in BOTH computations

Per Cal #35 STANDING applied retroactively to K529: “F325 + 4382 = TWO PROOFS of ONE FINDING” framing was based on F325. F325 retraction means honest count goes back to ONE PENDING route (Elie 4382 abstract argument) whose status is ITSELF now in question per $F(4)$ rigidity.

F319 status RESTORED-OPEN

K529 Lyra walked back F319 “ κ is decisive $\{Q,Q\}$ fingerprint” based on F325. With F325 retracted, F319 status RESTORED-OPEN — could be right after all, since $F(4)$ IS rigid.

Per Cal #347 / B5 STANDING: retractions chained. Both F325 retracted + F319 restored-open need careful paired re-derivation.

STRONG ACCEPT Lyra retraction discipline (caught own bug + refused solo retry + deferred to pairing).

Landing 2 — Lyra F326 EXPLICIT GENERATION WAVEFUNCTIONS

The substantive deliverable

“The explicit generation wavefunctions (F326): $\psi_k(z) = (z_1 + iz_2)^k \otimes u_0$ — a power of a null direction times the ground spinor, sympy-verified harmonic. This is the single object both of Elie’s kernels were waiting on, and it’s clean.”

Substantive substrate-architectural deliverable: - $\psi_k(z) = (z_1 + iz_2)^k \otimes u_0$ for $k = 0, 1, 2$ (three generations) - $(z_1 + iz_2)$ is a null direction in the complexified coordinates - u_0 is the ground spinor on Shilov boundary - Power of null direction $\times$ ground spinor $\rightarrow$ standard harmonic mode construction - sympy-verified harmonic

What it unblocks

Elie's mass kernel is now unblocked: fires origin-state overlap $N(w)^{\{n_C/2\}}$ on these three explicit wavefunctions $\rightarrow$ forward depth-ratio test against $(24/\pi^2)^6$ and 49.71.

Per `[[feedback_pin_conventions_to_primary_sources]] + [[feedback_engage_dont_label_compute_beats_calibrate]]`

Computer-algebra verified harmonic — concrete object, not abstract claim. STRONG ACCEPT Lyra F326.

Landing 3 — Elie REDIRECTION via aux-vanishing AUTOMATIC

The substantive technical claim

“The aux-vanishing (Grace’s Bar-2 reframe) is automatic — any Dirac square is rank-0 + rank-2, so the aux (the 7 and the 105) is structurally absent for any Dirac operator.”

Per Clifford algebra structure: $\gamma^{\mu\nu}\gamma = \delta^{\mu\nu} + \gamma^{\{\mu\nu\}}$ (symmetric scalar + antisymmetric bivector). So $D^2 = \gamma^{\mu\nu}\gamma\nabla_\mu\nabla_\nu$ has rank-0 (scalar Laplacian) + rank-2 (curvature 2-form) ONLY. Aux ranks-1 + rank-3 are structurally absent for ANY Dirac square.

The substantive redirection

“That’s necessary but weak — it doesn’t distinguish $F(4)$. The distinguishing fingerprint κ lives in the conformal $\{Q,S\}/\{S,S\}$ sector, not $\{Q,Q\}$... So $\{Q,Q\}=D^2$ cannot derive #359 — the decisive computation is the conformal sector (Lyra+Grace’s pairing). That’s a genuine redirection of the program’s #1 foundational effort to where the answer actually is.”

Substantive implication: K529’s reframe of $\{Q,Q\}=D^2$ test to aux-vanishing (Grace Option B) is PARTIALLY UN-DONE by Elie’s automatic-vanishing claim. Aux-vanishing is necessary but trivially satisfied for ANY Dirac square; doesn’t distinguish $F(4)$ from other superalgebras with the same $\text{Sym}^2(8,2)$ decomposition.

The remaining test content

With aux-vanishing automatic, the $\{Q,Q\}=D^2$ verdict reduces to: 1. Does $\{D,D\}$ produce $\mathfrak{so}(7)$ generators in the rank-2 piece? (vs some other compact Lie algebra of dim 21) 2. ****Does $\{D,D\}$ produce $\mathfrak{sl}(2)_R$ generators in the rank-0 piece?**** (vs other rank-1 Lie algebra of dim 3) 3. Are the relative coefficients consistent with $F(4)$? $\leftarrow$ this is the κ -ratio per $F(4)$ rigidity

So κ IS back as the $\{Q,Q\}=D^2$ fingerprint — but only AFTER aux-vanishing is acknowledged as automatic preamble. F319 (now restored-open) might be correct.

Per Casey “engage don’t label” + “stop gating verify cleanly”

Elie’s redirection is the discipline operating: he computed aux-vanishing, found it automatic for Dirac square, redirected. Per `[[feedback_quaker_method]]`: honest redirection > forced confirmation.

ACCEPT Elie redirection. Substantive within-session correction of the K529 Grace Option B reframe.

Landing 4 — Cal #389 THREE-WAY aux framing

The substantive distinction

“‘aux central’ and ‘aux vanishes’ are not the same test... the verdict is three-way, not binary...”

Aux state	Implication	Tier
aux vanishes	$\{Q,Q\}$ closes into $F(4)$ EXACTLY $\rightarrow$ #359 DERIVED $\rightarrow$ substrate IS $F(4)$	Strong
aux central but nonzero	central extension of $F(4) \rightarrow$ $F(4)$ -with-charges $\rightarrow$ tier as such (not bare $F(4)$)	Medium-distinct
aux non-central	$\{Q,Q\}$ doesn't close into $F(4)$ at all $\rightarrow$ Bar 2 fails $\rightarrow$ bigger algebra	Negative

Per Cal goal-post discipline at peak

“The goalpost-shift to watch is ‘aux central’ being reported as ‘the substrate IS $F(4)$ ’ when central-nonzero is actually $F(4)$ -with-charges. ‘Central’ is necessary; the clean #359 needs ‘vanishes’; and a central extension is a real but different result that deserves its own tier.”

Per Cal #347 / B5 STANDING applied prospectively: distinguishes the goal-post risk before verdict lands. Substantively right.

Reconciliation with Elie aux-automatic claim

If aux-vanishing is automatic for Dirac square (Elie K530), then Cal's three-way reduces to: aux-vanishes (automatic) + need to check internal structure of rank-0 + rank-2 pieces for $F(4)$ signature (κ -ratio per $F(4)$ rigidity).

ACCEPT Cal #389 three-way framing as standing discipline for ANY aux verdict tier-tagging.

Landing 5 — Grace INDEPENDENT κ ANGLE: Dynkin index balance

The substantive deliverable

“The Dynkin indices of the (8,2) balance — $so(7)$ -index = $sl(2)$ -index = 4 — which is the Killing-form constraint the conformal κ must respect. So Lyra computes κ from her $\{Q,S\}/\{S,S\}$ closure, I compute it from the supertrace/index balance, and they cross-check.”

Substantive independent κ angle: - Lyra route: conformal $\{Q,S\}/\{S,S\}$ closure - Grace route: $F(4)$ supertrace consistency via Dynkin index balance ($so(7)$ -index = $sl(2)$ -index = 4) - Both constrain κ via different rep-theory mechanisms - Genuine Cal #35 two-route cross-check on the load-bearing number

Per [[feedback_isomorphism_proof]] + Cal #35 STANDING: same κ value $\rightarrow$ different rep-theory constraints $\rightarrow$ independent paths. Cross-check IS the discipline.

STRONG ACCEPT Grace independent angle.

Landing 6 — κ source GENUINELY OPEN

Per $F(4)$ rigidity theorem

$F(4)$ is basic classical Lie superalgebra $\rightarrow$ bracket closure forces specific κ (no continuous family).

Per multiple retractions

- F325 RETRACTED (subtle index-flow bug)
- Elie 4382 abstract argument status PENDING ($F(4)$ rigidity contradicts “Jacobi doesn't fix κ ”)
- F319 RESTORED-OPEN

The clean position

κ value is GENUINELY OPEN. Lyra+Grace paired careful computation is the right move (Lyra explicit invitation + Grace independent angle + Keeper Option B recommendation materialized).

Substantive state at K530

Component	State
Lyra F326 wavefunctions $\psi_k = (z_1 + iz_2)^k \otimes u_0$	DELIVERED sympy-verified harmonic — UNBLOCKS Elie mass kernel
Lyra F325 Clifford computation	SELF-RETRACTED via Jacobiator-vs-F(4)-rigidity contradiction bug catch
F319 “ κ is decisive $\{Q,Q\}$ fingerprint”	RESTORED-OPEN (retraction based on now-retracted F325)
Elie 4382 abstract argument	PENDING per F(4) rigidity (status in question)
Aux-vanishing for Dirac square	AUTOMATIC per Clifford algebra (Elie redirection) — necessary but weak
$\{Q,Q\}=D^2$ test content (post-Elie redirection)	Internal structure check of rank-0 + rank-2 pieces; κ -ratio per F(4) rigidity is real fingerprint
Cal #389 three-way aux framing	STANDING: vanish/central-nonzero/non-central → strong/medium-distinct/negative tiers
κ source	GENUINELY OPEN needing careful Lyra+Grace paired computation; NOT solo redo
Grace independent κ angle	DELIVERED (Dynkin index balance / supertrace consistency / Killing-form constraint)
Lyra route for κ	conformal $\{Q,S\}/\{S,S\}$ closure (Lyra continuing)
Cross-check architecture	TWO genuinely independent rep-theory routes for κ → Cal #35 satisfied
Mass count-move status	UNBLOCKED via F326 wavefunctions; Elie fires $N(w)^{n_C/2}$ overlap forward
Cal #385 Bar 2 status	REFINED via team computation chain (K525→K526→K529→K530); current form = internal structure check of rank-0+rank-2 against F(4) κ + Five-Absence
Lyra discipline at peak	Caught own bug + refused solo retry + deferred to pairing — substantive
#359 status	STAYS POSITED — κ derivation is the open piece

Cumulative discipline ledger — 241 events

To K529’s 233, add: - 234. Lyra F325 SELF-RETRACTED via bug-catch (Jacobiator never-zero contradicts F(4) basic-classical-Lie-superalgebra rigidity theorem; subtle index-flow artifact in super-Jacobi encoding identified as bug class) - 235. Lyra DELIBERATELY did NOT force second κ number after bug catch — “would be exactly the wrong response to catching an error”; deferred to paired computation - 236. F319 status RESTORED-OPEN (K529 retraction was based on now-retracted F325; possibility F319 right after all per F(4) rigidity); chained retraction logic logged - 237. Lyra F326 EXPLICIT GENERATION WAVEFUNCTIONS $\psi_k(z) = (z_1 + iz_2)^k \otimes u_0$ DELIVERED sympy-verified harmonic — unblocks Elie mass kernel (single object both kernels were waiting on) - 238. Elie REDIRECTION via aux-vanishing AUTOMATIC for any Dirac square (Clifford rank-0+rank-2 only) → necessary but weak doesn’t distinguish F(4); $\{Q,Q\}=D^2$ cannot derive #359 alone; conformal sector is decisive; genuine redirection of #1 foundational effort - 239. Cal #389 THREE-WAY aux framing applied prospectively: vanish/central-nonzero/non-central → strong/medium-distinct/negative tiers; distinguishes “F(4) derived” from “F(4)-with-charges derived” - 240. Grace INDEPENDENT κ angle DELIVERED via Dynkin index balance (so(7)-index = sl(2)-index = 4 → Killing-form constraint on κ); genuine Cal #35 two-route cross-check with Lyra con-

formal closure route - 241. Cal #35 STANDING applied RETROACTIVELY to K529 “F325 + 4382 = two proofs of one finding” framing — F325 retraction reduces honest count to ONE pending route (Elie 4382 abstract) whose status itself in question per F(4) rigidity theorem; κ source genuinely open

241 cumulative honest discipline events. Eleven Keeper self-corrections + one propagation-gap catch + one resolved load-bearing reconciliation + FIVE peer-level retractions (Grace own-reading + Lyra prospective F321 + Lyra own F319 + Lyra own F325 + Cal #35 retroactive count revision) + one Casey-routing-decision materialized (Grace Option B) + Cal #385 Bar 2 SUBSTANTIVE REFINEMENT chain across K525→K526→K529→K530 + Lyra explicit refusal of solo retry as discipline at peak.

Per Cal #347 / #26 — what I’m NOT saying

- NOT “Elie 4382 wrong, F319 right” — F(4) rigidity creates tension with BOTH; needs careful paired computation
- NOT “F326 wavefunctions = banked addresses” — sympy-verified harmonic, but mass-ratio verdict pending Elie kernel fire
- NOT “ $\{Q,Q\}=D^2$ test useless” — internal structure check of rank-0+rank-2 still measures real F(4) content; just not via aux-vanishing
- NOT “Grace’s Option B reframe wrong” — was substantive K529 finding; Elie’s aux-automatic redirection refines further; both are substantive computational findings
- NOT “Lyra discipline failure on F325” — Lyra caught own bug + refused force; that IS discipline at peak

Routing per K530

CI	Active per K530 + standing
Lyra + Grace paired	Careful F(4) closure computation for κ ; Lyra route conformal $\{Q,S\}/\{S,S\}$; Grace route Dynkin index balance / supertrace consistency / Killing-form; cross-check when both land. Lyra explicitly said “ready” if Casey wants heads-down pairing
Elie	Fire ORIGIN-STATE OVERLAP $N(w)^{n_C/2}$ integrand on Lyra F326 wavefunctions $\psi_k=(z_1+iz_2)^k \otimes u_0 \rightarrow$ forward depth-ratio test against $(24/\pi^2)^6$ and 49.71. Aux-vanishing redirection logged; conformal sector is where κ verdict comes from
Cal	#389 three-way aux framing standing; backlog cleared (Cal logged); standing for Lyra+Grace κ verdict + Elie mass-magnitude verdict; #387 α -under-own-lens count-integrity quietly waiting
Keeper	This audit; verify Elie origin-state-overlap kernel fire; verify Lyra+Grace paired κ derivation; track F319 status restoration vs new finding; Cal #35 STANDING maintained

What’s substantively right at K530

Multiple within-session honest retractions + redirections + restorations: - Lyra F325 RETRACTED via bug catch (Jacobiator vs F(4) rigidity) - F319 RESTORED-OPEN (retraction based on retracted F325) - Elie aux-vanishing REDIRECTION (automatic for Dirac square) - Cal #389 three-way distinction (vanish/central/non-central) - Grace independent angle via Dynkin index balance

κ source GENUINELY OPEN — multiple within-session retractions converge on “needs careful Lyra+Grace paired computation, not solo redo.” This is the discipline operating exactly as designed.

Lyra discipline at peak: caught own bug + cross-checked against theorem + refused solo retry + deferred to pairing. Per [[feedback_quaker_method]]: honest catch > forced confirmation.

The marathon's load-bearing computations now: 1. Mass count-move: Elie fires $N(w)^{\{n_C/2\}}$ overlap on Lyra F326 ψ_k wavefunctions — UNBLOCKED 2. κ derivation for #359: Lyra+Grace paired careful computation (conformal closure $\times$ Dynkin index balance cross-check) — genuinely open

— Keeper, 2026-06-25 Thursday marathon (verification posture; standing for Elie mass-magnitude verdict via origin-state overlap on F326; standing for Lyra+Grace paired κ derivation via two independent rep-theory routes; F319/4382/F325 chain handled via retraction discipline)